# How to make justifiable inferences

# Presentation Outline

1. **How to make justifiable statistical inferences**
   - Populations and samples
   - Design- and Model-based inference
   - Two forms of randomization
     - Choice of units
     - Assignment of treatments
   - and four types of studies
2. **Examples**

# Population and Sample

Reminder:

$$\begin{array}{ccc} \text{Population} & \Longrightarrow & \text{Sample} \\ \downarrow & & \downarrow \\ \text{parameter} & \Longleftarrow & \text{statistic} \end{array}$$

# Design- and Model-based inference

Statistical inference is inference based on a probability model

- The probability model connects samples to the population
  - Design-based inference:
    - Estimation and inference justified only by study design,
    - I.e., only on how data collected.
  - Model-based inference
    - presume a model for the random mechanism
    - I.e., add assumption(s)

# Design-based: What is randomized?

Design-based inference

- What is randomized?
  - Choice of units: random, haphazard, or convenience / arbitrary
  - Assignment of treatments to units: randomly or not
- “units” are the items in the population
  - E.g., people, plots of land, nests, pots, vials
- The sample of units are those chosen for the study

# Choice of units

- How are the sample units chosen?
- Randomly: strict definition - by some formal probability mechanism
    - Random number table / computer program, Rolling dice
- Arbitrary / Convenience: The units that are available
    - Students in a class, participants in an online survey
    - Often not clear what population these are sample from
- Haphazard: Attempted to be random, but not formally
    - Choosing a plot of land by throwing a ball

# Assignment of treatments

- How are treatments assigned to units in the sample?
- Randomly: strict definition - by some formal probability mechanism
  - Same mechanisms as choice of units
  - Does not have to be independent assugnments
    - Random start systematic sample:
      A B C D A B C D or B C D A B C D A or C D A B C D A B or D A B C D A B C
- Haphazardly or Arbitrarily
  - Similar to choice of units
- Random assigment more common than random choice of units because easier
  - But what is the population when choice of units not random?

# Four types of studies

Four combinations of choice of units and assignment of treatments:
Display in Sleuth chapter 1

| | Treatment assignment | |
|---|---|---|
| Choice of units | Random | Not |
| Random | experiment (rare) | survey |
| Not | experiment | observation |

# Example - part 1

A study collected information on children at a middle school in the investigator's neighborhood. Kids at that school were classified by the number of hours per day they spent on social media. Children who spent more than four hours on social media each day were found to have higher cholesterol levels, on average, than children who spent less.

1. Are units selected at random or not? Be able to justify your choice.
2. Are units assigned to groups at random or not? Be able to justify your choice.
3. What is the most appropriate name for this study:
   Survey, observational, experimental (rare) and experimental?
   Be able to justify your choice.

# Example - part 2

A second study was very careful to randomly sample 1000 children from a list of all students attending middle schools in a large city. Also classified kids by amount of social media use and compared cholesterol levels.

1. Are units selected at random or not? Be able to justify your choice.
2. Are units assigned to groups at random or not? Be able to justify your choice.
3. What is the most appropriate name for this study:
   Survey, observational, experimental (rare) and experimental?
   Be able to justify your choice.

# Our answers

Example 1:

1. No: This is arbitrary / convenience choice of units (kids)
2. No: These are pre-existing groups
3. Observational

Example 2:

1. Yes: The kids in the study are a random sample from a well-defined population
2. No: Groups are still pre-existing
3. Survey, but better would be comparative survey because comparing 2 groups.

# Important points:

- Design-based and model-based inference
- Randomization of subjects vs. randomization of treatments
- 4 types of studies